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THE GENERAL VARIATIONAL PRINCIPLE OF TRANSPORT THEORY[1]

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1. INTRODUCTION

The solution of the Boltzmann equation is sometimes the most difficult part of the calculation of the transport properties of a physical system. In general, this is an integral equation over many variables; the solution often proceeds best by inspired guesswork rather than by well-justified analytical arguments. However, the pure mathematical theory of integral equations does provide a general theorem linking these equations with certain variational problems. Just as in the theory of partial differential equations, this theorem provides a formal method of writing down a sequence of approximate solutions converging to the correct answer. The method is of general validity, and with sufficient effort can give a practical solution to any given problem, although the labor of setting up the machinery and computing an answer is not always justified, especially when the details of the scattering processes are not known with great accuracy. The calculations of Kohler (1948, 1949*a*) and of Sondheimer (1950), yielding corrections to the Bloch theory of electrical conductivity of metals, are by this method, as also is the derivation by Howarth and Sondheimer (1953) of formulae for mobility in polar semiconductors.

It is usual to consider the variational method merely as a convenient mathematical device. Such an attitude robs it of most of its interest for the average physicist, who would probably prefer the whole discussion to be put in small type in an appendix. The purpose of this review is to rescue the principle from being labelled "mathematical technique", and to show that it has a considerable, easily grasped, "physical" content. Indeed, it is hoped to show that understanding of the physical interpretation of the principle can lead to more sophistication in its practical application to the solution of Boltzmann's equation in particular cases, and to some interesting generalizations which are not obvious in the purely mathematical formulation.

The key to this interpretation may be found in the paper by Kohler (1948), where it is shown that the "variational function", that is, the integral expression whose stationary property is required by the principle, is proportional to the rate of production of entropy by collision processes. But "entropy production" plays a central role in an entirely different branch of theoretical physics—the macroscopic theory of the thermodynamics of irreversible processes. The connection between the microscopic statistical theory, starting from Boltzmann's equation, and the macroscopic thermodynamic theory, where the parameters of the system are supposed known, is usually through the transport coefficients—the electrical and thermal conductivities, etc. This link is not always very firm, and may even be broken by mistake, as Sond-

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heimer (1956) has pointed out. The appearance in both theories of the same scalar quantity suggests a much more intimate relation, which should be made explicit. One recalls that it is very convenient to define the connection between *equilibrium* statistical mechanics and thermodynamics by the identification of entropy; Boltzmann's H-theorem then appears as a proof (or demonstration) of the Second Law of Thermodynamics. A variational theorem for entropy production in *non-equilibrium* statistical mechanics might thus be expected to correspond, in the same way, to a general law of steady state thermodynamics, and hence have a much greater content than as a mere mathematical theorem.

A discussion along such general lines may seem out of place in a symposium on the special topic of electron transport in solids. The excuse for opening this window on to a wider landscape is partly to look outward, to see where we are on the map, and partly to let the light inward, to show more clearly how to do the things we want to do within our own room. Arising out of detailed work in the special field of electron transport, the investigation leads us away into realms of abstract theory, but in the end we return to the particular problem with some new tools and new modes of thought.

2. THE VARIATIONAL PRINCIPLE

We shall attempt to avoid too abstract a formulation by talking in terms of a simple standard problem—the electrical conductivity of a metal containing static impurity centers. For completeness, we define a standard notation, following Wilson (1953). Boltzmann's equation is deduced from the balancing of the rates at which electrons are induced to change from one quasi-stationary state (labelled by the wave vector $\mathbf{k}$) to another. An electric field $\mathscr{E}$ causes $f_{\mathbf{k}}$, the distribution function for electrons in state $\mathbf{k}$, to change at the rate

$$\left.\frac{\partial f_{\mathbf{k}}}{\partial t}\right]_{\text{drift}} = -\frac{e}{\hbar}\,\mathscr{E}\cdot\nabla_{\mathbf{k}} f_{\mathbf{k}} \approx -e\,\mathscr{E}\cdot\mathbf{v}_{\mathbf{k}}\frac{\partial f_{\mathbf{k}}^{0}}{\partial E_{\mathbf{k}}} \tag{1}$$

if $f_{\mathbf{k}}$ is close to its equilibrium value $f_{\mathbf{k}}^{0}$. Here $\mathbf{v}_{\mathbf{k}}$ is the group velocity, and $E_{\mathbf{k}}$ the energy, in state $\mathbf{k}$.

On the other hand, suppose that the probability of an electron, known to be in state $\mathbf{k}$, being scattered into state $\mathbf{k}'$ (known to be initially empty) lying in the volume element $\mathbf{dk}'$ in k-space, is $Q(\mathbf{k}, \mathbf{k}')\mathbf{dk}'$. Then the collision rate of change of $f_{\mathbf{k}}$ is given by

$$\left.\frac{\partial f_{\mathbf{k}}}{\partial t}\right]_{\text{scatter}} = -\int \{f_{\mathbf{k}}(1-f_{\mathbf{k}'})Q(\mathbf{k}, \mathbf{k}') - f_{\mathbf{k}'}(1-f_{\mathbf{k}})Q(\mathbf{k}', \mathbf{k})\}\,\mathbf{dk}'. \tag{2}$$

We take account here of the exclusion principle.

In (2) we have distinguished explicitly between $Q(\mathbf{k}, \mathbf{k}')$ and $Q(\mathbf{k}', \mathbf{k})$ which is the measure of the inverse scattering process, from $\mathbf{k}'$ to $\mathbf{k}$. According to the principle of microscopic reversibility, these two functions are equal, i.e.

$$Q(\mathbf{k}, \mathbf{k}') = Q(\mathbf{k}', \mathbf{k})\,. \tag{3}$$

This symmetry is an essential point in what follows.

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It is convenient to define a new function, $\Phi_{\mathbf{k}}$, which is a measure of the deviation of $f_{\mathbf{k}}$ from equilibrium. We write

$$\begin{aligned} f_{\mathbf{k}} &= f_{\mathbf{k}}^0 - \Phi_{\mathbf{k}}\, \partial f_{\mathbf{k}}^0/\partial E_{\mathbf{k}} \\ &= f_{\mathbf{k}}^0 + \Phi_{\mathbf{k}} f_{\mathbf{k}}^0(1-f_{\mathbf{k}}^0)/kT \end{aligned} \tag{4}$$

by direct differentiation of $f_{\mathbf{k}}^0$ which is a Fermi function depending only on $E_{\mathbf{k}}$.

Adding (1) and (2) to get the total rate of change of $f_{\mathbf{k}}$, which must be zero in the steady state, and using (3) and (4), we obtain the Boltzmann equation in the following form:

$$-e\,\mathscr{E}\cdot\mathbf{v}_{\mathbf{k}}\frac{\partial f_{\mathbf{k}}^0}{\partial E_{\mathbf{k}}} = \frac{1}{kT}\int (\Phi_{\mathbf{k}}-\Phi_{\mathbf{k}'})\, P(\mathbf{k}, \mathbf{k}')\, \mathbf{dk}'. \tag{5}$$

In this formulation we must define a new probability function,

$$P(\mathbf{k}, \mathbf{k}') = f_{\mathbf{k}}^0(1-f_{\mathbf{k}'}^0)Q(\mathbf{k}, \mathbf{k}'), \tag{6}$$

which gives the scattering rate, in the equilibrium state, between $\mathbf{k}$ and $\mathbf{k}'$, that is, subject to the probability, $f_{\mathbf{k}}^0$, of $\mathbf{k}$ being actually occupied and, $(1-f_{\mathbf{k}'}^0)$, of $\mathbf{k}'$ being unoccupied, at any given moment. This function is also symmetric, i.e.

$$P(\mathbf{k}, \mathbf{k}') = P(\mathbf{k}', \mathbf{k})\,. \tag{7}$$

This follows directly from (3), remembering that energy is conserved in the process so that $f_{\mathbf{k}}^0 = f_{\mathbf{k}'}^0$. Alternatively, (7) is merely a special case of the principle of detailed balance.

Some care has been taken here to establish a notation which will be suitable for later discussion of more complex cases, although it is perhaps a little more formal than is necessary for the present simple problem.

In principle, we know the left-hand side of (5) and the kernel $P(\mathbf{k}, \mathbf{k}')$; the Boltzmann equation is thus an inhomogeneous integral equation for the unknown distribution function $\Phi_{\mathbf{k}}$. It belongs to a class of equations which may be represented in the simplest form by

$$\begin{aligned} X(k) &= \int \{\Phi(k)-\Phi(k')\}\, P(k, k')\, dk' \\ &= L\Phi \end{aligned} \tag{8}$$

where $X(k)$ is a known function and $P(k, k')$ is a known symmetric kernel. To apply the variational principle, one other fact is needed; $P(k, k')$ must always be of the same sign. This is a natural property of (6) since $P(\mathbf{k}, \mathbf{k}')$ is a probability, and hence always positive. We suppose, therefore, that $P(k, k')$ is positive definite in (8).

For compactness of notation we have introduced the symbol L for the integral operator in (8), acting on Φ. We also introduce the convention

$$\langle \Phi, \Psi \rangle \equiv \int \Phi(k)\,.\,\Psi(k)\, dk \tag{9}$$

which is equivalent to defining an "inner product" of the functions.

The variational principle may now be stated. There are a number of different forms, of which the following is typical (e.g. Wilson 1953):

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I. Let Φ be the solution of the integral equation (8) and Ψ any function for which

$$\langle \Psi, L\Psi \rangle = \langle \Psi, X \rangle . \tag{10}$$

Then, of all functions Ψ satisfying this relation, Φ is the one which makes $\langle \Psi, L\Psi \rangle$ a maximum.

The proof is quite simple. In its barest terms, we have

$$\begin{aligned} \langle \Psi, L\Phi \rangle &\equiv \iint \Psi(k)\, P(k, k')\, \{\Phi(k) - \Phi(k')\}\, dk\, dk' \\ &= \iint \Psi(k')\, P(k, k')\, \{\Phi(k') - \Phi(k)\}\, dk\, dk' \end{aligned} \tag{11}$$

from the symmetry of $P(k, k')$. Adding gives

$$\begin{aligned} \langle \Psi, L\Phi \rangle &= \frac{1}{2} \iint \{\Psi(k) - \Psi(k')\}\, P(k, k')\, \{\Phi(k) - \Phi(k')\}\, dk\, dk' \\ &= \langle \Phi, L\Psi \rangle \end{aligned} \tag{12}$$

from the symmetry of the integral as between Φ and Ψ.

Moreover, putting $\Phi = \Psi$, we obtain an integral with a square and a positive definite function in the integrand, i.e., for any Ψ

$$\langle \Psi, L\Psi \rangle \geqslant 0. \tag{13}$$

Condition (10) and the fact that Φ satisfies the integral equation (8) give

$$\langle \Psi, L\Phi \rangle = \langle \Psi, X \rangle = \langle \Psi, L\Psi \rangle . \tag{14}$$

From (13) we have

$$\begin{aligned} 0 &\leqslant \langle (\Phi - \Psi), L(\Phi - \Psi) \rangle \\ &= \langle \Phi, L\Phi \rangle + \langle \Psi, L\Psi \rangle - \langle \Phi, L\Psi \rangle - \langle \Psi, L\Phi \rangle \\ &= \langle \Phi, L\Phi \rangle + \langle \Psi, L\Psi \rangle - 2\langle \Psi, L\Phi \rangle, \text{ from (12)}, \\ &= \langle \Phi, L\Phi \rangle - \langle \Psi, L\Psi \rangle, \qquad \text{from (14)}, \end{aligned}$$

i.e.

$$\langle \Phi, L\Phi \rangle \geqslant \langle \Psi, L\Psi \rangle, \tag{15}$$

which we are required to prove.

The theorem is used as follows. We suppose that Φ may be expanded in a series of the form

$$\Phi(k) = \sum_n a_n \phi_n(k) \tag{16}$$

where $\phi_n(k)$ are known functions of k, making up, perhaps, a complete sequence, whilst the variables a_n are supposed to be chosen later to make Φ the solution of the integral equation. But substituting into the integral expressions $\langle \Phi, L\Phi \rangle$ and $\langle \Phi, X \rangle$ yields formulae containing integrals of the form $\langle \phi_n, L\phi_m \rangle$ and $\langle \phi_n, X \rangle$ which are coefficients of quadratic and linear forms in a_n. The principle is then applied to give the best choice of a_n, by differentiation with respect to these variables, making the variational function a maximum subject to the subsidiary condition (10). This yields a set of linear equations, which can be solved by simple algebra. Of course, we

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need an infinite sequence to obtain an exact solution of the integral equation, but by careful choice of the functions $\phi_n(k)$ we may usually get a good approximation to the answer with only a few terms. Crudely speaking, the art of guessing the functional form of the solution of the equation is refined and extended, at the cost of a certain amount of algebra and arithmetic, in a systematic way. From the solution thus obtained, the transport coefficients may be calculated in the usual manner.

3. ENTROPY PRODUCTION

Having stated the principle in its simplest form, as a mathematical theorem, we now consider the physical approach. If we look back from (8) to (5), that is, from the basic simplified integral equation to the Boltzmann equation of which it is a model, we see that the analogue of the variational function (i.e., in the condensed notation, $\langle \Phi, L\Phi \rangle$) must be the expression obtained by multiplying the right-hand side of (5) by $\Phi_{\mathbf{k}}$ and integrating with respect to $\mathbf{k}$. In other words, the variation process is to be applied to

$$(17) \quad \frac{1}{kT} \int \Phi_{\mathbf{k}} \int (\Phi_{\mathbf{k}} - \Phi_{\mathbf{k}'}) \, P(\mathbf{k}, \mathbf{k}') \, \mathbf{dk} \, \mathbf{dk}' = - \int \Phi_{\mathbf{k}} \frac{\partial f_{\mathbf{k}}}{\partial t}\Big]_{\text{scatter}} \mathbf{dk}.$$

But let us consider the statistical definition of entropy (e.g. ter Haar, 1954) in a Fermi–Dirac assembly, assuming without proof that this is valid in non-equilibrium states:

$$(18) \qquad S = -k \int \{ f_{\mathbf{k}} \ln f_{\mathbf{k}} + (1 - f_{\mathbf{k}}) \ln (1 - f_{\mathbf{k}}) \} \, \mathbf{dk}.$$

Let this equation be differentiated with respect to time, and let us assume that we are near equilibrium, so that $f_{\mathbf{k}}$ is close to $f_{\mathbf{k}}^0$ and the formula need be expanded only to the lowest power of $\Phi_{\mathbf{k}}$. A little manipulation yields the following:

$$(19) \qquad \frac{\partial S}{\partial t} = -\frac{1}{T} \int \Phi_{\mathbf{k}} \frac{\partial f_{\mathbf{k}}}{\partial t} \mathbf{dk} + \frac{1}{kT^2} \int E_{\mathbf{k}} \frac{\partial f_{\mathbf{k}}}{\partial t} \mathbf{dk}.$$

The second term may be dropped, since it is proportional to the rate of change of the total energy of the system, and this is zero for the type of collisions we are considering. In the steady state, of course, the over-all rate of change of $f_{\mathbf{k}}$ is also zero, so that the distribution is stationary and has constant entropy. But if we consider separately the two equal and opposite contributions to this rate of change, from drift and collisions respectively, we may infer two corresponding contributions to the production of entropy, i.e. for the scattering terms,

$$(20) \qquad \frac{\partial S}{\partial t}\Big]_{\text{scatter}} = -\frac{1}{T} \int \Phi_{\mathbf{k}} \frac{\partial f_{\mathbf{k}}}{\partial t}\Big]_{\text{scatter}} \mathbf{dk}.$$

Reference back to (17) shows that, apart from a constant factor $1/T$, this is just the variational function that we proposed for the Boltzmann equation. By the basic symmetry of $P(\mathbf{k}, \mathbf{k}')$, and with the same argument as for (12), we may write this expression in the particularly elegant form

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$$\left.\frac{\partial S}{\partial t}\right]_{\text{scatter}} = \frac{1}{2kT^2}\iint (\Phi_{\mathbf{k}}-\Phi_{\mathbf{k}'})^2 P(\mathbf{k},\mathbf{k}')\,\mathbf{dk}\,\mathbf{dk}'. \tag{21}$$

This is the result proved by Kohler. From it we see that the effect of collisions is always to increase the entropy, in agreement with the Second Law of Thermodynamics.

In addition to the integral $\langle\Psi, L\Psi\rangle$ in (10), there also occurs the expression $\langle\Psi, X\rangle$. It will be remembered that $X(\mathbf{k})$ is the known function on the left of the integral equation; from the way we derived the Boltzmann equation, we see that the analogue of $\langle\Phi, X\rangle$ is just

$$\begin{aligned} -\frac{1}{T}\int \Phi_{\mathbf{k}}\, e\mathscr{E}\cdot \mathbf{v}_{\mathbf{k}}\frac{\partial f_{\mathbf{k}}^{0}}{\partial E_{\mathbf{k}}}\,\mathbf{dk} &= \frac{1}{T}\int \Phi_{\mathbf{k}}\left.\frac{\partial f_{\mathbf{k}}}{\partial t}\right]_{\text{drift}}\mathbf{dk} \\ &= -\left.\frac{\partial S}{\partial t}\right]_{\text{drift}}, \end{aligned} \tag{22}$$

that is, minus the rate at which entropy changes as a result of drift. In fact, since (22) must equal (21), the effect of the electric field is to reduce the entropy in the material, tending always to produce more order in the electron distribution.

It is interesting to verify that (22) actually does represent entropy production in macroscopic terms. From (22) and (4) we have

$$\begin{aligned} -\left.\frac{\partial S}{\partial t}\right]_{\text{drift}} &= \frac{1}{T}\mathscr{E}\cdot\int e\mathbf{v}_{\mathbf{k}}(f_{\mathbf{k}}-f_{\mathbf{k}}^{0})\,\mathbf{dk} \\ &= \frac{1}{T}\mathscr{E}\cdot\mathscr{J}, \end{aligned} \tag{23}$$

where $\mathscr{J}$ is the electric current* in the steady state. This is exactly what we should write for the production of entropy in the form of Joule heat, for the current $\mathscr{J}$ flowing down the electric field $\mathscr{E}$ at temperature T. We may look upon the whole irreversible transport process as the production of entropy, at rate (21), by scattering inside the material, and the subsequent removal of the same amount of entropy (leaving the material in the steady state of constant entropy) by the electric field, sweeping it into the reservoirs which maintain isothermal conditions, where it appears in precisely the amount we should calculate from macroscopic thermodynamics.

4. THERMODYNAMIC FORMULATION

In the theory of the thermodynamics of irreversible processes, entropy production also enters, according to the following formula (e.g. de Groot 1951):

$$\frac{\partial S}{\partial t} = \sum_k X_k J_k \qquad (k = 1, 2, \ldots n, \text{ say}) \tag{24}$$

which defines the generalized thermodynamics "forces", X_k, conjugate to a given set of generalized "currents", J_k. In order to interpret the variational

*Strictly speaking, it is the current *density* multiplied by the volume, but we shall assume throughout that we are dealing with a unit cube, in which there is no difference between current and current density, entropy and entropy density, etc.

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principle in this language, we must find entities to which we may give these names. It is clear that the distribution function, $\Phi_{\mathbf{k}}$, is the analogue of a current, for by summation with suitable scale factors, as in (23), over all the states $\mathbf{k}$, it yields the actual macroscopic current. We may think of $\Phi_{\mathbf{k}}$ as measuring the actual, net, out-of-balance flux of electrons in the $\mathbf{k}$th state, just as if it were the current, J_k, in the kth branch of an electrical circuit. Correspondingly, from (22) and (24), the generalized force conjugate to $\Phi_{\mathbf{k}}$ must be

$$X_{\mathbf{k}} = -\frac{1}{T} e \mathscr{E} \cdot \mathbf{v}_{\mathbf{k}} \frac{\partial f_{\mathbf{k}}^0}{\partial E_{\mathbf{k}}}. \tag{25}$$

It is only necessary to think of the integral over $\mathbf{k}$ as a sum, which it strictly is in a finite specimen, to make the correspondence complete.

Let us first translate the Boltzmann equation, (5), into this language. We may write it, for the present purpose, in the form

$$X_{\mathbf{k}} = \int L(\mathbf{k}, \mathbf{k}') \, \Phi_{\mathbf{k}'} \, d\mathbf{k}' \tag{26}$$

where the kernel $L(\mathbf{k}, \mathbf{k}')$ is compounded in an obvious way out of $P(\mathbf{k}, \mathbf{k}')$, etc., and has the same symmetry and sign properties. This is equivalent to writing

$$X_k = \sum_{k'} L_{kk'} J_{k'}, \tag{27}$$

i.e., *the relationship between the forces and currents is linear, with the coefficients* $L_{kk'}$. Moreover, from the properties of $L(\mathbf{k}, \mathbf{k}')$ we may infer that *the coefficients* $L_{kk'}$ *form a positive definite symmetric matrix.* These statements are none other than the principles proposed by Onsager, as the foundation of the standard thermodynamic theory. The symmetry relations

$$L_{kk'} = L_{k'k} \tag{28}$$

are the famous Onsager relations. It is clear that our interpretation of the steady state is consistent with this essential principle.

Now to the main business, which is to find a rule for deducing these relations. Look again at (21), which defines the entropy production due to collisions. This is a quadratic function of $\Phi_{\mathbf{k}}$. A given set of currents forced through an array of scattering centers or other dissipative devices gives rise to a definite amount of entropy, independently of how those currents are created. We write

$$\left.\frac{\partial S}{\partial t}\right]_{\text{scatter}} = \sum_{kk'} L_{kk'} J_k J_{k'} = \left.\frac{\partial S}{\partial t}\right]_{\text{in}} \tag{29}$$

to show the quadratic dependence, using (26) and (27). We call this the *intrinsic*, or *internal* entropy production, because it depends only on the local properties, local currents, etc.

On the other hand, we already have formula (24), which corresponds to (22) and (23). We call this the *extrinsic* entropy production

$$-\left.\frac{\partial S}{\partial t}\right]_{\text{drift}} = \sum_k X_k J_k = \left.\frac{\partial S}{\partial t}\right]_{\text{ex}} \tag{30}$$

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since it can be calculated from measurements made outside the system, determining the forces and currents, and does not depend on the local behavior of the electrons nor on the way in which they are scattered.

But we have seen that (29) and (30) correspond to the functions introduced in the statement and proof of the variational principle I, that is, to $\langle \Phi, L\Phi \rangle$ and $\langle \Phi, X \rangle$ in (10). Gathering together the various threads, we construct the following statement of the principle:

II. Consider all distributions of currents such that the intrinsic entropy production equals the extrinsic entropy production for the given set of forces. Then, of all current distributions satisfying this condition, the steady state distribution makes the intrinsic entropy production a maximum.

The proof for the Boltzmann equation goes without further logic than the interpretation of the language into the formalism of principle I. The same proof implies that (27) may be derived from the same principle, and from the definitions (29) and (30). The symmetry relation (28) is an obvious consequence of the fact that the intrinsic entropy production (29) is *defined* as a quadratic form in the current variables, and therefore automatically has symmetric coefficients.

It will be seen, however, that we have now what appears to be a general principle of the thermodynamics of irreversible processes. Put briefly, we say that, in a system subject to fixed forces, the currents will distribute themselves so as to make the entropy production by internal dissipative processes a maximum, subject only to the condition that this must equal the production of entropy estimated from external measurements of these currents and forces. The actual result we have deduced from a special case, but the statement II is in general terms. The fact is that this is a known general principle (although for some reason it is not given in any of the books on the subject). It was deduced by Onsager (1931) from general statistical arguments, and was precisely the basis of his derivation of the Onsager relations (28).

In a restricted form, the principle is much older. For example, it is closely related to Rayleigh's theorem involving the dissipation function (Rayleigh 1896). Perhaps the most familiar case is the theorem concerning the distribution of currents, J_k, in a network of resistances $L_{kk'}$ with given applied electromotive forces, X_k. This requires that the currents be such that

$$\sum_{kk'} L_{kk'} J_k J_{k'} - 2 \sum_k J_k X_k \tag{31}$$

is a minimum (Jeans 1920). The principle, II, as we have written it, is not quite the same as this, although it is mathematically equivalent. For example, in (31) there is a function to be *minimized*, whereas II refers to the *maximum* of the entropy production. If we look at (31), we see that it corresponds to

$$\frac{dS}{dt}\Big]_{\text{in}} - 2\frac{dS}{dt}\Big]_{\text{ex}}, \tag{32}$$

which is, in the steady state, just the *negative* of the actual production of entropy in the dissipation process. This is also the form of the theorem stated by Onsager.

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An alternative statement, also easily proved (e.g. Wilson 1953), is that we must minimize

$$\sum_{kk'} L_{kk'} J_k J_{k'} \Big/ \left(\sum_k J_k X_k \right)^2 = \frac{dS}{dt}\Big]_{\text{in}} \Big/ \left\{ \frac{dS}{dt}\Big]_{\text{ex}} \right\}^2 \tag{33}$$

with respect to the currents. This expression reduces to the *inverse* of the actual entropy production in the steady state, so that a minimum might be expected. The form of the principle given in I and II seems the most natural from a physical point of view, though not necessarily the most concise mathematical formulation.

The various formulations have been noted explicitly, because there is danger of confusion with another theorem where entropy production is extremal. This is Prigogine's theorem, which states, crudely speaking, that if not all the forces acting on a system are fixed the free forces will take such values as to make the entropy production a minimum. Taking for granted the linear relations (22) between forces and currents, it tells us what to expect when we let go of some of the leads, and allow the potentials to seek their own levels. The two theorems seem to be entirely independent of one another.

5. PRACTICAL ADVANTAGES OF THE GENERAL PRINCIPLE

Although there is some amusement in unmasking the mathematical theorem, and denouncing it as a mere "physical principle", our real business is to show that this approach to the problem has practical advantages in the actual calculation.

In the first place, entropy and entropy production are good physical quantities, more or less easily defined for any given system. It is thus an easy matter to find the special variational principles in detailed mathematical terms, for Boltzmann equations of any degree of complexity. For example, consider the case of phonon–electron scattering, as in the Bloch model, but allowing for some deviation from equilibrium in the phonon distribution. If there is also a thermal gradient ∇T, we must solve the following equations:

$$\begin{cases} -e\mathscr{E}\cdot \mathbf{v_k} \dfrac{\partial f_{\mathbf{k}}^{\,0}}{\partial E_{\mathbf{k}}} - \mathbf{v_k}\cdot \nabla T \dfrac{\partial f_{\mathbf{k}}^{\,0}}{\partial T} = \dfrac{1}{kT} \iint (\Phi_{\mathbf{k}} + \Phi_{\mathbf{p}} - \Phi_{\mathbf{k}'})\, P_{\mathbf{kp}}^{\ \mathbf{k}'}\, \mathbf{dk'dp} \\ \qquad\qquad + \dfrac{1}{kT} \iint (\Phi_{\mathbf{k}} - \Phi_{\mathbf{p}} - \Phi_{\mathbf{k}'})\, P_{\mathbf{k}}^{\mathbf{pk}'}\, \mathbf{dk'dp}; \\ -\mathbf{v_p}\cdot \nabla T \dfrac{\partial N_{\mathbf{p}}^{\,0}}{\partial T} = \dfrac{1}{kT} \iint (\Phi_{\mathbf{k}} + \Phi_{\mathbf{p}} - \Phi_{\mathbf{k}'})\, P_{\mathbf{kp}}^{\ \mathbf{k}'}\, \mathbf{dk'dk}. \end{cases} \tag{34}$$

Here $N_{\mathbf{p}}^{\,0}$ is the equilibrium distribution function (Bose–Einstein) for phonons of wave vector $\mathbf{p}$. $\Phi_{\mathbf{p}}$ has much the same significance as $\Phi_{\mathbf{k}}$, except that the Bose–Einstein statistics give, instead of (4),

$$N_{\mathbf{p}} = N_{\mathbf{p}}^{\,0} + \Phi_{\mathbf{p}} . N_{\mathbf{p}}^{\,0}(1 + N_{\mathbf{p}}^{\,0})/kT \tag{35}$$

for the out-of-balance distribution.

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The probability function $P_{\mathbf{kp}}{}^{\mathbf{k}'}$ for the process by which electron $\mathbf{k}$ absorbs phonon $\mathbf{p}$ and goes into state $\mathbf{k}'$ is defined by

$$P_{\mathbf{kp}}{}^{\mathbf{k}'} = Q_{\mathbf{kp}}{}^{\mathbf{k}'} f_{\mathbf{k}}{}^{0} N_{\mathbf{p}}{}^{0}(1-f_{\mathbf{k}'}{}^{0}) = P_{\mathbf{k}'}{}^{\mathbf{kp}} \tag{36}$$

where $Q_{\mathbf{kp}}{}^{\mathbf{k}'}$ is the corresponding scattering probability when the initial states are known to be singly occupied and the final state is initially empty. The essential symmetry of $P_{\mathbf{kp}}{}^{\mathbf{k}'}$ for the forward and reverse processes may be proved in detail from the form of the distribution functions, from conservation of energy, and from microscopic reversibility (which imposes the same symmetry on $Q_{\mathbf{kp}}{}^{\mathbf{k}'}$). Here also, the principle of detailed balance gives an equivalent proof.

In calculating the entropy production, it is necessary to include the contribution of the phonons, which comes out, just as in (20), to

$$-\frac{1}{T}\int \Phi_{\mathbf{p}} \frac{\partial N_{\mathbf{p}}}{\partial t}\,\mathbf{dp}. \tag{37}$$

Using the symmetry properties of $P_{\mathbf{kp}}{}^{\mathbf{k}'}$ we find, altogether,

$$\left.\frac{\partial S}{\partial t}\right]_{\text{in}} = \frac{1}{\mathrm{k}T^2}\iiint (\Phi_{\mathbf{k}}+\Phi_{\mathbf{p}}-\Phi_{\mathbf{k}'})^2 P_{\mathbf{kp}}{}^{\mathbf{k}'}\,\mathbf{dkdpdk}', \tag{38}$$

which must be the variational function of our problem. No doubt this formula might have been guessed, but it is not immediately obvious by inspection of (34). It is easy to suggest any further generalizations, for scattering processes involving more and more particles.

Equation (38) provides a convenient starting point for the calculation of the effect of the phonon unbalance, which is discussed by Sondheimer in another paper in this series. It also represents, as it happens, the simplest approach to the original Bloch calculation, where $\Phi_{\mathbf{p}}$ is taken to be zero. The standard approach to this (e.g. Wilson 1953) is through the Bloch integral equation, derived from (34) by the assumption that

$$\Phi_{\mathbf{k}} = c(E_{\mathbf{k}}).k_z \tag{39}$$

where k_z is the component of $\mathbf{k}$ in the field direction and $c(E_{\mathbf{k}})$ is a function to be determined later by the solution of the integral equation.

The Bloch equation, as all will know who have attempted to use it, is extremely complicated, running to several lines of type in its simplest formulation. In the end, it is necessary to use the variational method to obtain an exact solution (Sondheimer 1950), introducing as a set of trial functions for $c(E)$ the various powers of E. We might just as well have used the corresponding set of trial functions for $\Phi_{\mathbf{k}}$ in the Boltzmann equation itself, with the variational function (38). It turns out that the integrals are much easier to evaluate, and the same solutions may be found with considerably less work. One can see, for example, that it is not necessary to count separately the processes in which a phonon is emitted and those in which it is absorbed; these have the same elementary probability, and correspond merely to forward and reverse directions of the process $\mathbf{k} + \mathbf{p} \rightleftarrows \mathbf{k}'$. Again, because $\mathbf{k}$, $\mathbf{k}'$, and $\mathbf{p}$ all

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appear on the same footing in the integrals, one may carry out the integration in whatever order is most convenient. The easiest way is to remove the integral over $\mathbf{p}$ by conservation of momentum, then to integrate over the energy of $\mathbf{k}$ using energy conservation, next to integrate over $E_{\mathbf{k}'}$ using the properties of the Fermi functions, and finally to consider the various orientations of the wave vectors $\mathbf{k}$ and $\mathbf{k}'$ on the Fermi surface. This order of integration is impossible in the Bloch integral equation, all of whose complexity arises from treating the angular variation of $\mathbf{k}$ much too early.

It would take us too far afield to demonstrate these results in detail. It is sufficient to say that the method has been used to derive the standard formulae for a wide range of problems in the theory of metals and semiconductors, and in every case the algebraic work is much shorter than in the conventional derivations. These include the corrections to the Bloch formulae (Sondheimer 1950), the extension to include Umklapp processes (Ziman 1954), inelastic electron scattering in rare earth metals (Elliott 1954) and its contribution to thermal resistance and thermoelectric properties, scattering of carriers by optic modes in polar semiconductors (Howarth and Sondheimer 1953), and the contribution of scattering by dislocations to the electrical resistance of a metal (Mackenzie and Sondheimer 1950; Hunter and Nabarro 1953).

The moral is that entropy is a scalar physical quantity, which may be defined in a very general way irrespective of the choice of representation. Entropy production, also, cannot depend on any special way of counting collision processes, nor on the choice of lattice directions for axes, and so on. That is, it has the properties of a mathematical invariant of the problem; it is the fixed center about which various coordinate systems and representations pivot. By using these properties explicitly, we have the opportunity, at the last possible moment, of choosing any special frame of reference which will make the final stages of the calculation as simple as possible.

6. EXPLOITING THE LINK WITH THERMODYNAMICS

Another practical advantage of the general method is that we may avoid a number of routine steps in obtaining the transport coefficients from the solution of the Boltzmann equation, and write down directly some general formulae without special models. This method is used on several occasions by Kohler and his co-workers. To illustrate the technique, we shall consider the problem of the conduction of heat by phonons in an insulating solid.

In general we must take account of two types of scattering process—direct scattering by impurities, imperfections, etc., and phonon–phonon interactions (Peierls 1955) in which $\mathbf{p} + \mathbf{p}' \rightleftarrows \mathbf{p}''$. The former we may suppose to go with probability $P(\mathbf{p}, \mathbf{p}')$ and the latter with probability $P_{\mathbf{pp}'}^{\mathbf{p}''}$, these quantities being defined just as in the previous cases. It is obvious that the entropy production corresponding to the distribution $\Phi_{\mathbf{p}}$ is given by

$$\left.\frac{dS}{dt}\right]_{\text{in}} = \frac{1}{2kT^2}\iint(\Phi_{\mathbf{p}}-\Phi_{\mathbf{p}'})^2 P(\mathbf{p},\mathbf{p}')\,\mathbf{dpdp}' + \frac{1}{kT^2}\iiint(\Phi_{\mathbf{p}}+\Phi_{\mathbf{p}'}-\Phi_{\mathbf{p}''})^2 P_{\mathbf{pp}'}^{\mathbf{p}''}\,\mathbf{dpdp}'\mathbf{dp}''. \tag{40}$$

(We should be careful to allow for the various polarizations, but this is trivial.)

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Now, $\Phi_{\mathbf{p}}$ corresponds to a distribution in which there will be a thermal current flowing, of amount

$$\mathbf{U} = -\int E_{\mathbf{p}}\mathbf{v}_{\mathbf{p}}\Phi_{\mathbf{p}}\frac{\partial f_{\mathbf{p}}^{0}}{\partial E_{\mathbf{p}}}\,\mathbf{dp}. \tag{41}$$

This current flowing down the temperature gradient ∇T represents a dissipative process in which entropy is produced at the rate

$$\left.\frac{dS}{dt}\right]_{\text{ex}} = -\nabla T\cdot\mathbf{U}/T^{2}$$
$$= U^{2}/\kappa T^{2} \tag{42}$$

where κ is the thermal conductivity of the medium (assumed isotropic). In the steady state, the entropy production by collisions equals the extrinsic macroscopic production, so that (40) and (42) are equal. We have at once from (41)

$$\frac{1}{\kappa} = \tag{43}$$

$$\frac{\frac{1}{2kT^{2}}\iint(\Phi_{\mathbf{p}}-\Phi_{\mathbf{p}'})^{2}P(\mathbf{p},\mathbf{p}')\,\mathbf{dp}\,\mathbf{dp}'+\frac{1}{kT^{2}}\iiint(\Phi_{\mathbf{p}}+\Phi_{\mathbf{p}'}-\Phi_{\mathbf{p}''})^{2}P_{\mathbf{pp}'}^{\mathbf{p}''}\,\mathbf{dp}\,\mathbf{dp}'\,\mathbf{dp}''}{\left\{-\frac{1}{T}\int E_{\mathbf{p}}\mathbf{v}_{\mathbf{p}}\Phi_{\mathbf{p}}\frac{\partial f_{\mathbf{p}}^{0}}{\partial E_{\mathbf{p}}}\,\mathbf{dp}\right\}^{2}}$$

for the value of the thermal resistivity.

Of course, this formula is meaningless unless $\Phi_{\mathbf{p}}$ represents a true stationary state, and is a solution of the Boltzmann equation; we cannot avoid that part of the calculation. However, the variational principle may be used here to very practical advantage. The denominator of (43) is $(U/T)^{2}$. But, from (41), we may write

$$U/T^{2} = \left.\frac{dS_{1}}{dt}\right]_{\text{ex}} \tag{44}$$

where the right-hand side means "the entropy production when the current U flows down the thermal gradient $\nabla T = 1$". Thus (43) is a ratio of the form

$$\frac{1}{\kappa} = \frac{1}{T^{2}}\left.\frac{dS}{dt}\right]_{\text{in}} \Big/ \left\{\left.\frac{dS_{1}}{dt}\right]_{\text{ex}}\right\}^{2} \tag{45}$$

which is nearly the same as occurs in (33), except that we choose the particular value unity for the thermal gradient and are thus interested in definite values for the thermodynamic forces.

It was remarked that the variational principle asserts that in the steady state the ratio (33) takes its minimum value. Thus, (43) is not merely an expression for $1/\kappa$ to be used when we have found $\Phi_{\mathbf{p}}$ by other means; it is a quantity which is always greater than or equal to the true thermal resistivity, the closeness of $1/\kappa$ depending on the closeness of $\Phi_{\mathbf{p}}$ to the true distribution. If we use the "inner product" notation, and consider the ratio in the form

$$\langle\Phi, L\Phi\rangle/\{\langle\Phi, X\rangle\}^{2}, \text{ with } X = 1, \tag{46}$$

we are reminded of the variational principle in quantum mechanics, in the form that the expectation value of the energy in any state ϕ, i.e.

$$\int\phi^{*}\mathfrak{H}\phi d\tau/\int\phi^{*}\phi d\tau, \tag{47}$$

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is always greater than or equal to the energy of the ground state of the system, the difference depending on the degree to which ϕ approximates to the ground-state wave function. We may think of the resistivity as the "expectation value" of the scattering operator L in the non-equilibrium state Φ, suitably normalized. The rule that the value of the energy estimated from (47) is always better than the approximation in the choice of ϕ will also hold in the present argument.

This formulation is particularly useful because in many cases we use just a single trial function to establish an approximate result, without bothering to compute a complete solution. In that case we shall know at once the sign of the error, and (43) contains the absolute minimum number of integrals that need to be evaluated, with no further algebraic steps to write down the transport coefficients. More complicated cases (for example, thermoelectric power in metals) can be represented similarly, and explicit formulae for the exact values of the transport coefficients (that is, using a complete sequence of trial functions, as in (16)) can be written down in terms of matrices with elements of the form $\langle\phi_n, L\phi_m\rangle$, etc. The employment of such general formulae for the transport coefficients avoids the danger, pointed out by Sondheimer (1956), of making inconsistent approximations in the calculation of the separate coefficients. For example, the Onsager relations are automatically satisfied.

7. A FORMULA IN THE THEORY OF LATTICE CONDUCTION

We have considered the case of phonon–phonon interaction because one can derive a formula of considerable interest and utility whose proof seems very difficult except by the present argument. We therefore digress for a moment from the general discussion, to give this derivation.

It is well known (Peierls 1955) that a system in which there are only phonon–phonon processes in which momentum is conserved (N-processes) would have infinite thermal conductivity. This follows at once from (43), by writing

$$\Phi_{\mathbf{p}} = \mathbf{p}\cdot\mathbf{u} \tag{48}$$

where $\mathbf{u}$ is some arbitrary vector. Suppressing $P(\mathbf{p}, \mathbf{p}')$ for the moment, we obtain in the denominator only

$$\frac{1}{kT^2}\iiint\{(\mathbf{p}+\mathbf{p}'-\mathbf{p}'')\cdot\mathbf{u}\}^2 P_{\mathbf{pp}'}^{\mathbf{p}''}\,\mathbf{dp}\,\mathbf{dp}'\mathbf{dp}'' \tag{49}$$

which vanishes if $\mathbf{p}+\mathbf{p}'=\mathbf{p}''$. Hence, (48) must be the solution of the Boltzmann equation, since it makes the ratio for $1/\kappa$ take its lowest possible value, that is, zero.

But now suppose that there are other scattering processes, as assumed in (43). If these are not very effective compared with the N-processes, it is reasonable to choose, as a first approximation trial function, just the same expression (48). The contribution of N-processes still vanishes, but there will be a contribution by impurities, etc., proportional to

$$\frac{1}{2kT^2}\iint\{(\mathbf{p}-\mathbf{p}')\cdot\mathbf{u}\}^2 P(\mathbf{p}, \mathbf{p}')\,\mathbf{dp}\mathbf{dp}'. \tag{50}$$

A Debye model for the phonon modes, and the assumption that $P(\mathbf{p}, \mathbf{p}')$ does

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not depend on the absolute orientation of the vectors $\mathbf{p}$ and $\mathbf{p}'$, yields the following formula for the thermal resistivity:

$$W_i = \frac{3}{c^2} \int \frac{c_p}{v_p l_p} dp \tag{51}$$

where c_p is the contribution to the total specific heat, c, of the modes of wave vector p (i.e. $c = \int c_p dp$); l_p is the mean free path of these same phonons for this type of scattering, i.e.

$$\frac{1}{l_p} = \frac{1}{v_p} \int (1 - \cos\theta)\, Q_p(\theta)\, d\Omega \tag{52}$$

where $Q_p(\theta)d\Omega$ is the probability that a phonon of wave vector p is scattered through the angle θ into the solid angle $d\Omega$.

In the approximation we are using, W_i is additive for various types of scattering, so that Matthiessen's rule is assumed. The formula is extremely useful, because it is easy to calculate and offers no convergence difficulties. It has been used for various calculations, for example the scattering of phonons by electrons in semiconductors (Ziman 1956) and by isotopic mixtures (Berman, Foster, and Ziman, to be published). It is not, of course, exact, and gets worse as the scattering effect gets larger and N-processes no longer dominate the distribution function. In that case we may take it only as an upper bound to the true resistivity, and must calculate more deeply.

Finally, the effect of Umklapp processes (U-processes) can also be estimated. For these, (49) does not vanish, but contains just the square of $\mathbf{K} \cdot \mathbf{u}$ where $\mathbf{K}$ is a reciprocal lattice vector. This is exactly the method used by Leibfried and Schlömann (1954) to calculate W_U, the contribution of the U-processes to the thermal resistivity of a dielectric solid.

8. PHENOMENA IN A MAGNETIC FIELD

The discussion so far has been mainly a review of the ideas implicit in Kohler's original paper, and mention of the various applications of the method that have actually been made. We now introduce an entirely new and in some respects puzzling problem, which is at present being studied by Mr. F. Garcia-Moliner and the author.

Suppose that to our original model of electrons scattered by static obstacles we add a magnetic field $\mathbf{H}$. If the field is not too large to upset our original quantization scheme, the Boltzmann equation (5) acquires an extra term on the left, i.e.

$$\left.\frac{\partial f_{\mathbf{k}}}{\partial t}\right]_{\text{mag}} = -\frac{e}{\hbar} (\mathbf{v}_{\mathbf{k}} \wedge \mathbf{H}) \cdot \nabla_{\mathbf{k}} f_{\mathbf{k}}. \tag{53}$$

The substitution (4) for $f_{\mathbf{k}}$ gives no term from $f_{\mathbf{k}}^0$, so that we must use the full expression. Eliminating unnecessary complexity by the symbolism of (26), we may write the Boltzmann equation schematically in the form

$$X_{\mathbf{k}} = \int L(\mathbf{k}, \mathbf{k}')\, \Phi_{\mathbf{k}'} d\mathbf{k}' + M(\mathbf{H})\Phi_{\mathbf{k}} \tag{54}$$

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where the "magnetic scattering operator" is defined by

$$M(\mathbf{H}) = \frac{e}{\hbar T}\frac{\partial f_{\mathbf{k}}^{0}}{\partial E_{\mathbf{k}}}(\mathbf{v}_{\mathbf{k}}\wedge\mathbf{H})\cdot\nabla_{\mathbf{k}}. \tag{55}$$

The general effect of the magnetic field thus appears similar to that of the scattering obstacles. The electrons are deflected as they come into contact with the magnetic lines of force. There is an important difference; in magnetic deflection all the electrons are bent round together, and the ordering of the distribution is not changed directly. This is made clear if we calculate the entropy produced by the magnetic field, using (20) and (53). We write

$$\begin{aligned}\left.\frac{dS}{dt}\right]_{\text{mag}} &= \int \Phi_{\mathbf{k}}\, M(\mathbf{H})\ \Phi_{\mathbf{k}}\,\mathbf{dk} \\ &= \frac{e}{\hbar T}\int \Phi_{\mathbf{k}}\frac{\partial f_{\mathbf{k}}^{0}}{\partial E_{\mathbf{k}}}(\mathbf{v}_{\mathbf{k}}\wedge\mathbf{H})\cdot\nabla_{\mathbf{k}}\Phi_{\mathbf{k}}\,\mathbf{dk}.\end{aligned} \tag{56}$$

But use of the standard theorems for the manipulation of $\nabla_{\mathbf{k}}$ separates this into a "surface" integral, which vanishes because $\Phi_{\mathbf{k}}$ is certainly zero for sufficiently large $\mathbf{k}$, and other integrals which also vanish because they involve $(\mathbf{v}_{\mathbf{k}}\wedge\mathbf{H})\cdot\mathbf{v}_{\mathbf{k}}$ and $\nabla_{\mathbf{k}}\wedge\mathbf{v}_{\mathbf{k}}$. In fact, we find that (56) is identically zero: *there is no explicit contribution from magnetic scattering to the production of entropy.*

Now we apply the variational principle in the general thermodynamic form II, taking $\Phi_{\mathbf{k}}$ as a generalized current. Since $M(\mathbf{H})$ does not contribute to the entropy production, its effect cannot be felt in the calculation and it might as well never have been present. That is, we should find for the steady state not the solution of the Boltzmann equation (54) but *the solution of the equation without magnetic terms at all.* It is well known, of course, that the magnetic terms do have an effect, modifying the solution and hence altering the transport coefficients. Yet application of a general, well-established theorem of thermodynamics (or, at least, one whose truth has not been challenged for 20 years) seems to lead to a contradictory result.

For those familiar with the thermodynamics of irreversible processes, the fact that there should be difficulties in the case of a magnetic field will come as no surprise. It is well known, for example (de Groot 1951), that the Onsager relations (28) must then be modified to read

$$L_{kk'}(\mathbf{H}) = L_{k'k}(-\mathbf{H}) \tag{57}$$

showing that the coefficients are not symmetric, but are only made to agree by reversal of the magnetic field. A similar situation holds where there are Coriolis forces.

We have taken some trouble in §4 to show that Boltzmann's equation may be interpreted into thermodynamic language, and that it is consistent with the grammar of that language. Now in place of (27) we write

$$\begin{aligned}X_k &= \sum_{k'} L_{kk'}(\mathbf{H})\ .\ J_{k'} \\ &= \sum_{k'}\{L^{0}_{kk'}+M_{kk'}(\mathbf{H})\}\ .\ J_{k'}\end{aligned} \tag{58}$$

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where $L^0_{kk'}$ represents the scattering operator without magnetic terms and $M_{kk'}(\mathbf{H})$ is a matrix representation of the magnetic collision operator, $M(\mathbf{H})$. From the vanishing of the entropy production for magnetic scattering (which we may write $\sum_{kk'} J_k M_{kk'}(\mathbf{H}) J_{k'}$), we find

$$\begin{aligned} M_{kk'}(\mathbf{H}) &= -M_{k'k}(\mathbf{H}) \\ &= M_{k'k}(-\mathbf{H}) \end{aligned} \tag{59}$$

since $M(\mathbf{H})$ is linear in $\mathbf{H}$. Thus, the coefficients $L_{kk'}(\mathbf{H})$ in (58) satisfy the proper Onsager relations, (57), in the presence of a magnetic field. The Boltzmann equation is thus represented quite consistently by (58), and the thermodynamic language is intelligible.

But the paradox concerning the application of the general variational principle is still unresolved. Indeed, the rot has spread, since the lack of symmetry of the transport coefficients in the magnetic field leads to failure of Prigogine's theorem also. The antisymmetric parts (cf. (59)) of the coefficients $L_{kk'}(H)$ do not contribute explicitly to entropy production, and disappear from the calculation at the first step, although they will necessarily be required at the end in the equations defining the steady state. This point seems to have been overlooked in the usual treatments.

Let us look at the problem from another aspect. Does there exist any function which can be used to fix the final state by a variational principle? We return to a purely mathematical treatment; mention of the antisymmetry of the matrix $M_{kk'}(\mathbf{H})$ suggests an answer. In proving the original theorem, I, we have used explicitly the fact that the operator L is real and symmetric (more technically, self-adjoint). But the eigenvalue theory of real symmetric matrices can be generalized to include complex matrices if these have Hermitian symmetry, i.e. if

$$L_{kk'} = L^*_{k'k} \tag{60}$$

where $L^*_{k'k}$ means "the complex conjugate of $L_{k'k}$". The possibility of talking about antisymmetric matrices has thus been allowed, since this is the property of the imaginary part of the Hermitian matrix $L_{kk'}$.

The connection between the eigenvalue theory and the variational principle is very close. It is easy to prove that equations of the type

$$X_k = \sum_{k'} L_{kk'} J_{k'}, \tag{61}$$

where $L_{kk'}$ is a complex Hermitian matrix and the variables J_k may be complex, can be derived from variation of the Hermitian form

$$\sum_{kk'} J_k^* . L_{kk'} . J_{k'}, \tag{62}$$

subject to suitable subsidiary conditions.

To exploit this analogy, we need something equivalent to complex conjugation. In (60) this has the effect of reversing the sign of the antisymmetric part of the operator $L_{kk'}$. The corresponding effect in (59) is achieved by reversing the direction of the magnetic field $\mathbf{H}$. We propose the notation

$$M^*(\mathbf{H}) = M(-\mathbf{H}); \tag{63}$$

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the conjugate of the operator M is the same operator with the magnetic field reversed in direction. In the same way, we may define a distribution function $\Phi_{\mathbf{k}}^*$ which is a solution of the conjugate equation, i.e. of the Boltzmann equation

$$X_{\mathbf{k}} = \int L(\mathbf{k},\mathbf{k}')\Phi_{\mathbf{k}'}^* \, \mathbf{dk}' - M(\mathbf{H})\Phi_{\mathbf{k}}^*. \tag{64}$$

With this notation, it is easy to construct a general variational principle. In effect, we find that *the general argument leading to the principal I is maintained intact if we adopt the convention that, in the presence of a magnetic field,*

$$\langle \Phi, \Psi \rangle \equiv \int \Phi^*(k)\, \Psi(k)\, dk. \tag{65}$$

Detailed derivation of this result, and its application to problems of magnetoresistance, etc., will be given elsewhere. One can, for example, write down conditions that a linear combination of trial functions should be the best approximation to the solution. It turns out that these are just the conditions proposed by Kohler (1949*b*) without formal proof. Here we have a theorem which guarantees their status, and shows, for example, that the corresponding estimates of resistivity must be greater than, or equal to, the true resistivity.

The most interesting question still remains: What now is the relation between the variational principle and the thermodynamic formulation II? We have seen that the unmodified thermodynamic principle does not lead to the correct answer. However, the new variational function,

$$\langle \Psi, L\Psi \rangle = \int \Psi_{\mathbf{k}}^* \{L_0 + M(H)\}\, \Psi_{\mathbf{k}} \mathbf{dk}, \tag{66}$$

has still some meaning, in that, when $\Psi = \Phi$ and we are talking of the steady state distribution, (66) equals the actual production of entropy by collisions. In general, (66) is a function of the state of the system differing from the entropy production in that state, but the two functions become the same in the steady state. The effect of the magnetic terms is felt in fixing the steady state, but once that is reached these terms do not contribute explicitly to the production of entropy.

An alternative and, in my opinion, more fruitful convention is to suppose that the thermodynamic principle, II, is always true, in the general form in which we have stated it, and to modify the formula for the entropy production so that it always equals the variational function. That is, in place of (20) we write

$$\frac{dS}{dt} = -\frac{1}{T} \int \Phi_{\mathbf{k}}^* \cdot \frac{\partial f_{\mathbf{k}}}{\partial t} \mathbf{dk}, \tag{67}$$

which might be supposed to arise from some generalized formula for the entropy, in place of (18). (It is easy to find such a formula, but it is not unique and we shall not discuss it here.)

This assumption is not so radical as one might at first suppose. Entropy is only properly defined in the equilibrium state, and it is only tentatively that

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one extends the definition to non-equilibrium assemblies. We have seen that the two definitions coincide in the steady state; it is when the distributions are varying with time in a magnetic field that we find a difference, and this is a situation for which the old statistical arguments are certainly not adapted.

The failure of Prigogine's theorem is particularly interesting, because there we employ only macroscopic variables. But the same device, of postulating a conjugate current and then of writing (62) for the entropy production, works there also. Another case where something similar seems to be necessary occurs when we consider the effects of boundaries or inhomogeneities in the medium. This problem is being studied by Mr. S. Simons at the Cavendish Laboratory.

The heart of the matter seems to be the failure of the principle of microscopic reversibility in the presence of a magnetic field. In order to retrace the motion, we must not only reverse velocities but also the directions of all magnetic fields. If there were an argument giving an unambiguous combinatorial definition of entropy in non-stationary states, might it not depend on the conditions for time reversal?

Setting aside this very deep question, the difficulty at the moment seems to be in giving a clear meaning to Φ_k^*, which is the conjugate to a distribution function. The best we can do seems to be as follows: Suppose that Φ_k is an arbitrary distribution in the given magnetic field. One could imagine a set of forces for which Φ_k is actually the steady state solution. Then Φ_k^* is the steady state distribution with those same electric fields, etc., but with the magnetic field reversed. This is clumsy, but at least has "physical" significance.

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DISCUSSION OF ZIMAN'S PAPER

There was no time for detailed discussion, but **Prigogine** gave notice that he doubted the validity of part of Ziman's thermodynamic interpretation. **Ziman** agreed that further discussion on this point would be useful.

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